Design to Increase Social Interaction between Viewers of Dramas on OTT Platforms

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ABSTRACT

In this paper, we examined the effect of Over-the-Top (OTT) media platforms on social interaction and the sense of connection among drama viewers. The findings from semi-structured interviews revealed that the personalized and asynchronous nature of OTT streaming could lead to reduced social communication opportunities before, during, and after drama-viewing experiences. In particular, participants expressed difficulties in finding companions to co-watch dramas, a lack of simultaneous emotional communication with others, and concerns about spoilers. To address these challenges, we proposed three solution ideas and evaluated them in follow-up interviews. Based on our findings, we discuss design implications for promoting social interaction between OTT drama viewers.

CCS CONCEPTS

• **Human-centered computing → Collaborative and social computing**.

KEYWORDS

Sense of connection, social interaction, drama-watching, OTT, over-the-top, video streaming service, co-watching

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1 RESEARCH BACKGROUND

Media consumption is not just an individual's activity but a social one as well, serving as a conduit for the exchange of ideas and emotions among communities and fostering a sense of connection [4, 12]. Levy and Windahl [6] saw that social interaction around content could occur ***before watching*** (e.g., choosing the drama to watch together), ***during watching*** (e.g., exchanging thoughts and emotions while watching drama together), and ***after watching*** (e.g., sharing reflections on drama). Research has shown that such interaction can increase engagement with the media content [8, 11]. Therefore, promoting social interactions around media content can be as important as media consumption itself [15].

Traditionally, watching TV dramas has been regarded as a communal experience among individuals in the same physical location. In TV-based drama-watching environments, viewers had limited options to choose from the programs scheduled on television channels [2]. Therefore, it was common for people to watch the same TV dramas simultaneously in a shared setting, such as a living room with their family members. These settings provided opportunities for discussion and exchanging thoughts both during and after watching dramas [6, 8, 14].

As opposed to conventional TV drama viewing environments, OTT (over-the-top) platforms allow users to consume content at any time at any place as long as they have access to the Internet. Also, OTT platforms also offer a wide selection of dramas based on algorithms tailored to individual viewers. Consequently, social interaction and a sense of connection associated with live TV drama watching may no longer be as prevalent with viewers viewing dramas on OTT platforms [1]. This shift by OTT platforms may have diminished the sense of connection between viewers and have changed the way of sharing viewing experiences with others.

In light of this, the primary objective of this study is to examine how the OTT environment affected social communication among drama viewers (**RQ1**) and to explore design solutions for enhancing social communication among OTT drama viewers (**RQ2**).

To address our RQs, we first conducted semi-structured interviews with 15 OTT drama viewers to understand their overall experience watching OTT dramas and their social interaction and sense of connection with others. Through semi-structured interviews, we

identified the social interaction needs of participants before, during, and after watching dramas on OTT platforms. Based on these needs, we developed solution ideas to promote social interaction among drama viewers on OTT platforms, which were then evaluated and given feedback by 10 participants. In conclusion, we offer design implications for enhancing the OTT watching experience by facilitating social communication among OTT drama viewers.

2 RESEARCH METHOD

To address RQ1, we first conducted semi-structured interviews to understand participants' overall experience watching dramas on OTT platforms and their sense of connection with others. We recruited participants who had watched at least three dramas on OTT platforms within three months of the study to capture their vivid experiences and opinions. Fifteen participants who met the selection criteria above were recruited through online drama communities (10 females, average age: 25.3 years old). All research procedures were conducted after the IRB approval from the university where the study was hosted.

Interview questions contained questions about participants' general drama-viewing experience on TV and OTT platforms, as well as their social interactions related to drama-watching. Interviews were conducted via Zoom, which took about 30-40 minutes per participant. Following the interview, each participant received a gift card worth 10,000 Korean Won (equivalent to $8 USD).

The transcripts of the interviews were then analyzed using thematic analysis [3]. Three researchers independently coded the interview transcripts, then compared each other's themes and codes to identify common themes. Through several rounds of discussion, we identified challenges that participants faced when they sought to engage in social communication with other OTT drama viewers. In particular, we identified three types of challenges participants encountered in social communication before, during, and after watching OTT dramas.

Based on these challenges, we brainstormed solution ideas to address their specific needs to increase social interaction between drama viewers on OTT platforms (see Figure 1). Three solution ideas were then envisioned as plug-ins for existing OTT platforms (e.g., Netflix) and were developed into system mockup images using Figma (see Figure 2-4).

Finally, a follow-up interview was conducted to receive feedback on our solution ideas. Among the 15 participants who participated in the semi-structured interviews, ten who could arrange a time for follow-up interviews participated. Following the introduction of each solution idea with the system mockup images (Figure 2-4), participants shared their thoughts about each solution idea, discussing its advantages and limitations, willingness to use, and suggestions for improvement. The follow-up interviews took about 20 minutes and were also conducted online via Zoom. A gift card worth 5,000 Korean Won (equivalent to $4 USD) was compensated to each participant. Using the same coding procedure as before, we analyzed follow-up interview transcripts.

3 FINDINGS (RQ1)

Analyzing the semi-structured interviews, we discovered that all participants valued having increased choice and control over when and what to watch on OTT platforms. Unlike the traditional TV drama-watching experience, participants reported that they mostly viewed dramas alone on their mobile devices through OTT platforms: "*I watch dramas by myself, 95% of the time*" (P2). Some mentioned they preferred watching dramas alone on OTT platforms as it helped them to fully immerse themselves in the drama: "*I prefer watching dramas alone so that I can fully immerse myself into the drama*" (P3).

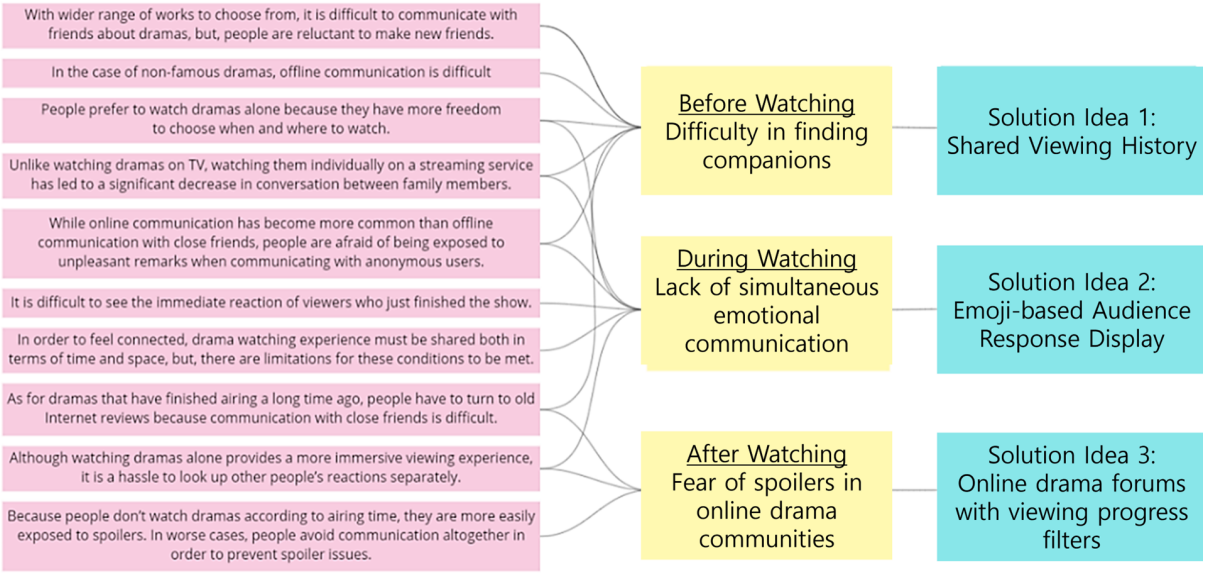

Figure 1: Thematic map of the findings from the data analysis

While participants were generally satisfied with the control and flexibility provided by OTT platforms, they desired some level of social interaction and a sense of connection with others that they had experienced in TV-based drama-watching experience. However, they reported challenges in initiating and engaging in such social interaction in OTT-based drama-viewing environments. The following sections outline three primary challenges in social communication among OTT drama viewers **before**, **during**, and **after watching OTT dramas**.

3.1 Obstacles to Interaction "Before Watching": Difficulty in Finding Companions

We found that most participants struggled to find companions who were watching the same drama concurrently. This resulted in fewer opportunities for real-time communication and sharing of viewing experiences with their acquaintances: "*I recently watched a drama that came out three years ago. Each episode was so much fun to watch, but it was too late to tell my friends about it*" (P7). Further, due to personalized recommendation algorithms within OTT platforms, they said that they tended to watch only dramas that suited their taste, making it challenging to find companions to share the drama-viewing experience with. In particular, participants who enjoyed dramas of a specific genre had trouble finding friends with similar interests: "*My friends don't watch Japanese dramas, so I don't have anyone to talk to*" (P7).

3.2 Obstacles to Interaction "During Watching": Lack of Simultaneous Emotional Communication

We found that participants occasionally wanted to observe the immediate emotional reactions of others while watching dramas. Although they generally believed that watching dramas alone helped them to concentrate more on each scene, they also expressed a desire to share emotional reactions with others during specific moments in the drama: "*When people are able to share each other's reactions to a drama as they are watching it together, they are able to experience the same emotions together, making the watching experience more engaging and memorable*" (P3).

Interestingly, despite the participants' desire for a co-watching experience, many were unaware of co-watching options available on OTT platforms (e.g., "Netflix party"). Even those who were aware of its existence mentioned that scheduling simultaneous co-watching with others was difficult and burdensome, so they ended up watching alone: "*It's troublesome to arrange a time to watch dramas with my friends*" (P7).

3.3 Obstacles to Interaction "After Watching": Fear of Spoilers in Online Drama Communities

As finding acquaintances to share drama-viewing experiences together was challenging, they gradually turned to online forums and communities where they could easily find viewers of certain dramas. However, some participants noted that this often increased the risk of being exposed to "spoilers": "*I checked online to see how others reacted to my favorite episode, but I also got to know what would happen in subsequent episodes through their comments*" (P9). Further, P3 stated that these concerns were almost inevitable when watching old dramas: "*If you search for information about old dramas, you will almost certainly find spoilers. So, you'd better avoid online communities if you do not want spoilers*" (P3).

Overall, our findings for RQ1 revealed three types of participants' social connection needs before, during, and after watching OTT dramas.

- **Before Watching**: I would like to find an acquaintance who can watch the drama I am interested in.
- **During Watching**: I watch dramas alone, at a time and place of my own choice. Yet, I also want to share my emotions with other viewers simultaneously.
- **After Watching**: I like browsing online forums for information and sharing experiences about dramas on OTT platforms, but I don't want to expose myself to spoilers.

4 SOLUTION IDEAS AND USER FEEDBACK (RQ2)

To address the need for social connection among OTT drama viewers, we brainstormed multiple ideas. Among them, we specifically developed three solution ideas into system mockups as plug-in applications that could be applied to existing OTT platforms like Netflix. Detailed descriptions and user interfaces of the solutions ideas are presented below, along with user feedback collected through follow-up interviews.

4.1 Solution Idea for the "Before Watching" phase: Shared Viewing History

In order to address the difficulty of finding acquaintances who watch the same drama, we proposed a solution idea that lets users add their offline friends and share their drama viewing history with them (see Figure 2). After users add friends, the shared viewing history feature allows users to see what dramas their friends are currently watching as well as what episodes they have watched up to. Additionally, based on each other's drama viewing history, it suggests the dramas to watch together. Finally, in case users do not wish to share their viewing records on some specific drama, they can selectively turn off the function.

In the follow-up interview, all participants evaluated this solution as highly helpful and likely to enhance communication and connections among close friends. They also said that they would be highly willing to use this solution. Many noted that it would help them quickly figure out and engage friends in a conversation about drama without having to ask them individually: "*This is helpful for me because I can easily bring up a topic if I know that someone watches a certain drama without having to ask about it*" (P13). In particular, P1, who enjoyed "*off-mainstream*" drama, highly evaluated the solution since "*repeatedly asking friends if they had seen the drama was burdensome.*" Other participants noted that seeing this shared list of viewing history would encourage them to choose the dramas their friends were watching.

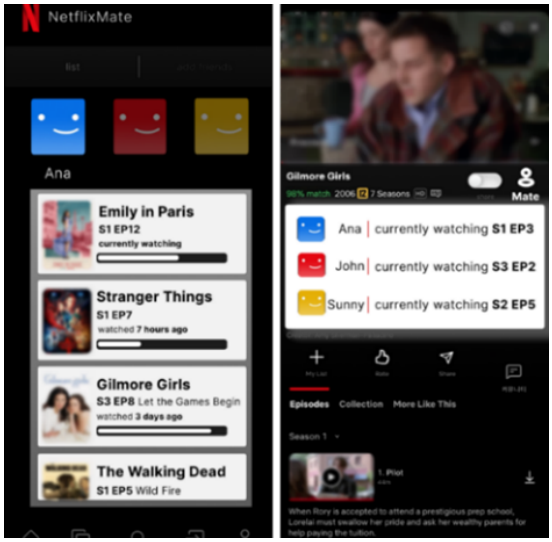

Figure 2: Mockup interface of the "Shared Viewing History" feature

4.2 Solution Idea for the "During Watching" phase: Emoji-based Audience Response Display

To create a shared emotional experience among viewers in the "During Watching" phase, we proposed a solution idea that allows viewers to share and observe each other's emotional reactions while they watch dramas alone (see Figure 3). By clicking the Emotions button on the video player, users can leave their emotional reactions using emojis for the specific moments of each episode. In addition, users can choose to display floating animated emojis that represent the dominant emotional reactions of other viewers. Since viewing and writing comments could greatly disrupt viewers' drama-watching experiences, emojis were chosen over written comments to promote simple and intuitive emotional interaction. This feature is similar to the live reactions feature offered by Instagram and Facebook during live streaming [7].

In follow-up interviews, most participants agreed that the proposed solution would address the issue of "lack of simultaneous communication with others" by allowing them to observe others' emotional reactions during drama watching, providing a sense of co-watching and social presence: *"It would help me feel like there are people watching with me at the same time"* (P4). Additionally, some participants thought the ability to record their emotional reactions using emojis would be helpful for archiving memorable scenes.

In spite of such general positive evaluation of the concepts of this solution, only a few participants expressed an interest in actually using it. Many participants were concerned that it would reduce their sense of immersion: "*With this feature, I might lose focus*" (P10). Further, P8 expressed concern about trolling, such as posting out-of-context responses to attract attention, when collecting real-time emotional responses from viewers: "*Imagine someone pressing the heart emoji during a scene where a character dies*" (P8).

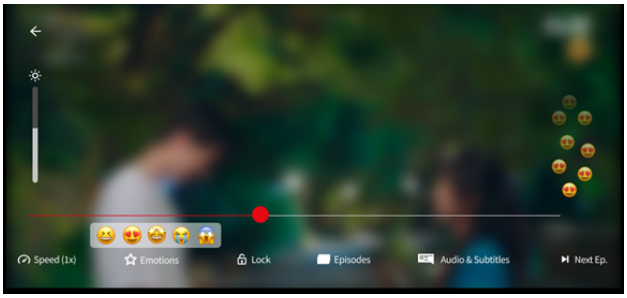

Figure 3: Mockup interface of the "Emoji-based audience response display" feature

4.3 Solution Idea for the "After Watching" phase: Online drama forums with viewing progress filters

The online drama forum with viewing progress filters (see Figure 4) was designed for users who do not want to be exposed to spoilers while being on online drama forums. In this solution, it keeps track of forum contributors' viewing progress as they write a forum post. Consequently, based on a user's viewing progress, the forum posts made by other users who watched or discussed subsequent episodes will automatically be filtered out. For example, those who have just finished watching episode 3 will not be exposed to posts from those who have watched episode 4 and beyond.

In the follow-up interview, many participants said that they were willing to use this solution. Specifically, participants who were initially reluctant to use online communities at all due to spoiler concerns gave positive feedback. "*I don't even search about things I haven't watched due to spoilers. As this can prevent spoilers, I see it as very useful*" (P15).

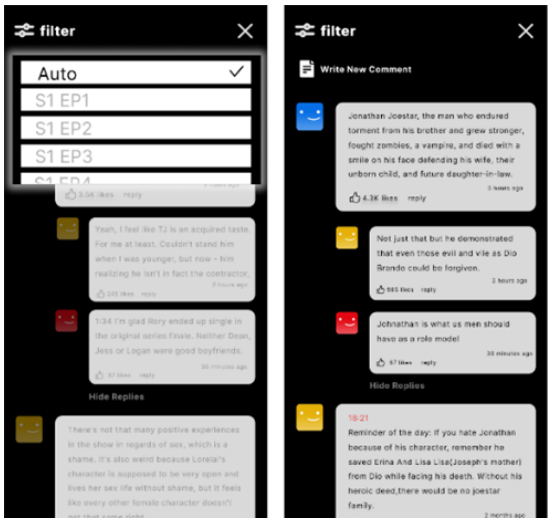

Figure 4: Mockup interface of the "Online drama forums with viewing progress filters" feature

5 DISCUSSION

In this study, we identified specific social communication challenges and needs that OTT drama viewers faced and suggested design solutions to address their challenges. Through semi-structured interviews, we found that the nature of OTT platforms could lead to decreased opportunities in offline communication about drama, as becoming difficult to find real-life connections to share drama-viewing experiences. Also, some reported that they often felt a lack of emotional communication in a simultaneous viewing experience and had concerns about spoilers in online forums. These challenges arise because OTT platforms allow people to watch dramas at different times, preventing people from aligning their viewing experiences.

In order to address social communication needs among viewers who are physically remote, many researchers have focused on simulating "co-watching" experiences. Co-watching refers to the act of watching a TV show or movie together with someone remotely in real-time, intending to foster a sense of connection among the viewers who are geographically apart [9]. Several co-watching applications have been proposed to allow people to watch videos together in synchronization, such as Amigo TV [5] and Zync [10]. Also, third-party services for OTT platforms such as Teleparty (previously Netflix Party) [13] facilitate co-watching experiences between friends and family members who are not co-located.

In general, prior studies found that co-watching experiences could help enhance the sense of connection among remote viewers [15]. Yet, it is uncertain whether solutions to facilitate co-watching alone could effectively address the specific social communication needs of OTT users, who place significant importance on the flexibility of watching dramas at their preferred time and in comfortable surroundings. To explore design space for facilitating social communication among OTT users while leveraging the benefits provided by OTT platforms, we introduced and evaluated three design solution ideas that could address the challenges that they encountered before, during, and after watching OTT dramas.

First, to address the challenges of finding acquaintances to share drama-viewing experiences, a feature that allows users to share viewing history was proposed. This feature may help users easily find acquaintances to share drama-viewing experiences, thus reducing the burden of initiating conversation. Specifically, we anticipate that this feature could facilitate the entire process of developing a sense of connection with offline connections, from choosing drama to viewing and sharing reflections. Next, to address the need for emotional communication in a simultaneous viewing experience, the "emoji-based audience response display" feature was proposed, which may allow users to feel a vivid sense of emotional connection with a large number of viewers. Finally, the "online forum with viewing progress filters" was proposed to meet the need for online communication free of spoilers. The feature may encourage users to participate more in online forums.

According to the study participants in the follow-up interviews, the two design solution ideas ("shared viewing history" and "online forum with viewing progress-based filters") received very positive evaluations, and participants expressed a high willingness to use them. However, while the concept of the "emoji-based audience response display" itself received positive feedback, only a few respondents were willing to actually use it. This indicates that the needs for communication and connection among viewers may not be the only elements that contribute to a satisfying drama-watching experience. Therefore, future technology aiming to enhance connection and communication among OTT drama viewers should take into account other factors, such as "immersion," "safety" and "flexibility."

6 CONCLUSION

In this study, we examined the challenges OTT viewers faced when engaging in social interaction during, before, and after watching OTT drama. In particular, participants reported difficulty finding companions to watch dramas with, a lack of simultaneous emotional communication with others, and concerns about spoilers in online drama forums. These challenges highlight the evolving nature of OTT platforms, where aligning viewers' drama-watching experience could not be easily achieved. As OTT platforms continue to be the main avenue for consuming video content, these challenges may become more prevalent, potentially leading to a more isolated watching experience over time.

To address these challenges, we proposed three solutions ideas to promote social interaction among OTT drama viewers: the "shared viewing history", "emoji-based audience response display", and "online drama forum with viewing progress filters." In the follow-up interview, participants gave positive feedback on the shared viewing history feature as a means to discover common drama interests with acquaintances. The online drama forum with viewing progress filters was also well-received, especially among those who had been cautious of spoilers. While the emoji-based audience response display was also viewed as an effective way to simulate shared real-time emotional communication, some raised concerns regarding potential distractions and the possibility of misusing this feature. This underscored the necessity for a balanced approach that takes into account not only the viewers' needs for connection and communication but also other important aspects, including immersion, safety, and flexibility.

Future research will be needed to develop working prototypes of solution ideas and conduct in-the-wild user studies to evaluate their applicability and effectiveness. Also, as this study primarily focused on drama-viewing experiences, a future study will need to investigate whether OTT viewers' needs and challenges for social communication would differ based on the genre of OTT video content (e.g., movie, documentary, reality show). Furthermore, it will be necessary to examine if our proposed solution ideas could be applicable to other types of OTT video content.

In conclusion, as social interaction and a sense of connection play a significant role in viewers' overall satisfaction with their drama-viewing experience, we hope our findings could contribute to enhancing the drama-watching experience on OTT platforms. We also wish that our insights and solution ideas could inspire other researchers and OTT platform designers to explore ways to keep OTT users connected while leveraging the benefits provided by OTT platforms.

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